Title page

Article type: Review

Title: Role of the Renin–Angiotensin System in Physiological and Pathological Cardiac Hypertrophy — Focus on the ACE2/Ang-(1–7)/Mas Axis

DECISÃO DO ORIENTADOR: o título é a decisão 5.8 do plano de corte e continua aberta. O corpo foi reorientado (a fração de palavras efetivamente sobre o eixo passou de ~28% na v1 para ~72% aqui), mas o título ainda lidera pelo assunto que o editor pediu para recuar. Ressubmeter com o título idêntico ao da versão recusada sinaliza que nada mudou. Opções: (a) manter; (b) inversão mínima — "The ACE2/Ang-(1–7)/Mas Axis in Physiological and Pathological Cardiac Hypertrophy" — que sinaliza o reenquadramento sem trocar o escopo; (c) o título da tese, "The ACE2/Ang-(1–7)/Mas Axis as a Regulator of the Quality of Cardiac Remodelling". O running head já lidera pelo eixo nas três opções. QUEM DECIDE: o Carlos, como orientador. O João chega com as três opções e a consequência de cada uma; a escolha é dele.

Running head: ACE2/Ang-(1–7)/Mas axis in cardiac hypertrophy

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[FALTA: só falta conferir contra o teto real de *review article* do JRAAS, que continua não verificado — as Submission Guidelines devolvem 403 a acesso automatizado. A contagem em si JÁ FOI MEDIDA com ferramenta em 04/08 e é 5.933, não estimativa: §1 460 · §2 303 · §3 383 · §4 2.201 · §5 353 · §6 1.194 · §7 304 · §8 527 · §9 208. Isso é 583 acima do alvo de 5.350 do plano de corte e 33 acima do teto da faixa de

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Abstract

The counter-regulatory ACE2/angiotensin-(1–7)/Mas axis of the renin–angiotensin system (RAS) appears to act in opposite directions in the two forms of cardiac hypertrophy: it restrains the maladaptive growth driven by the classical ACE/Ang II/AT_1 arm, yet it is required for the reversible, function-preserving growth of exercise. This review is centered on that axis and critically appraises the genetic and pharmacological evidence across experimental models, comparing findings between models rather than recounting them in sequence. We argue that the contradiction is apparent: the axis regulates the quality of remodeling rather than its magnitude — restraining pathological hypertrophy while being required for exercise-induced growth and, on thinner evidence, for that of pregnancy. We examine the interpretive limits that follow from the metabolic branch-point position of ACE2, the model dependence of pharmacological ACE2 activators, the discordance between endpoints of mass and endpoints of fibrosis, and the divergence between rodent and human RAS peptide dynamics in pregnancy. Throughout, we separate what we regard as established from what is emerging and what remains speculative, and we close by identifying the knowledge gaps and translational priorities that most constrain therapeutic targeting of the protective arm of the RAS.

[FALTA: confirmar o formato e o teto de palavras do resumo nas Submission Guidelines do JRAAS (a página `journals.sagepub.com/author-instructions/JRA` devolve HTTP 403 a acesso automatizado — pendência da seção 6 do plano de corte). Este resumo é não estruturado, seguindo a v2; se a revista exigir cabeçalhos estruturados para *review article*, ele precisa ser resegmentado antes da submissão]

Keywords: renin–angiotensin system; angiotensin-(1–7); ACE2; Mas receptor; cardiac hypertrophy; cardiac remodeling

1. Introduction

Cardiac hypertrophy is not a single entity. Because cardiomyocytes largely withdraw from the cell cycle shortly after birth and renew slowly thereafter, the adult heart accommodates sustained increases in workload chiefly by enlarging existing myocytes rather than by generating new ones.[1–3] The consequences of that growth, however, diverge sharply. Endurance exercise and pregnancy produce reversible growth that is matched by angiogenesis, spares the interstitium, and preserves or improves function; hypertension, valvular disease, and myocardial infarction produce growth accompanied by fibrosis, capillary rarefaction, metabolic and electrical remodeling, and a heightened risk of heart failure and arrhythmia.[1,4,5]

The renin–angiotensin system (RAS) is a principal neurohumoral determinant of which program prevails. Its classical arm — angiotensin-converting enzyme (ACE)/angiotensin II (Ang II)/AT_1 receptor — is an established driver of maladaptive remodeling, and its pharmacological blockade regresses left ventricular mass more than equivalent blood-pressure reduction achieved with other drug classes.[6,7] A counter-regulatory arm organized around angiotensin-converting enzyme 2 (ACE2), angiotensin-(1–7) [Ang-(1–7)], and the Mas receptor opposes these actions.[8,9]

That arm resists a simple antihypertrophic reading: it restrains hypertrophy in Ang II excess, hypertension, and myocardial infarction, yet it is required for the normal cardiac growth of exercise and, on thinner evidence, of pregnancy. We argue that this contradiction is apparent rather than real, and that it dissolves once the axis is understood to regulate not the magnitude of cardiac growth but its quality — the fibrotic, oxidative, metabolic, and angiogenic character of the remodeling that accompanies it. On this reading the axis is not a generic antihypertrophic brake but a determinant of whether growth follows an adaptive or a maladaptive program, and the therapeutic goal it implies is redirection of remodeling rather than suppression of growth.

The scope follows from this claim. We concentrate on the ACE2/Ang-(1–7)/Mas axis in cardiac hypertrophy, weighing genetic against pharmacological evidence and pathological against physiological models, and treating disagreement between models as informative rather than incidental. General cardiac cell biology, the wider RAS and the signaling of hypertrophy are covered only as far as is needed to situate the axis; we do not review the clinical pharmacology of ACE inhibitors and angiotensin receptor blockers, the renal and vascular actions of the protective arm, or the role of ACE2 as the SARS-CoV-2 entry receptor.[10] Alamandine/MrgD and alamandine-(1–5) enter only where they illuminate mechanisms shared across the protective arm.

This is a narrative and deliberately selective review, not a systematic one. Studies were identified in PubMed and Web of Science and selected for methodological informativeness — genetic as well as pharmacological manipulation, hemodynamic control, and separate reporting of mass and tissue endpoints — rather than for exhaustive coverage. The evidence base is overwhelmingly preclinical, and we mark throughout what we regard as established, what is emerging, and what remains speculative.

2. The renin–angiotensin system: classical and protective arms

The RAS cascade (Figure 1) begins when renin cleaves hepatic angiotensinogen to angiotensin I (Ang I), from which ACE removes a C-terminal dipeptide to generate the octapeptide Ang II, the ligand for the AT_1 and AT_2 G-protein-coupled receptors.[11–13] AT_1 mediates vasoconstriction, renal sodium and water retention, sympathetic facilitation, and aldosterone release, and its sustained activation drives hypertension, cardiac and vascular hypertrophy, fibrosis, and inflammation.[14,15] AT_2 generally opposes AT_1, eliciting vasodilatory, antiproliferative, anti-inflammatory, and antihypertrophic responses, although its contribution varies across diseases and remains incompletely defined.[16,17] Beyond the circulating cascade, the heart contains a complete local system capable of generating and degrading these peptides in situ, a distinction that recurs throughout this review because local and systemic signaling can diverge.[11]

The protective arm is organized around Ang-(1–7), generated chiefly by ACE2-mediated removal of the C-terminal phenylalanine of Ang II and secondarily from Ang I or angiotensin-(1–9) by neprilysin, prolyl endopeptidase, and ACE.[18–20] Two features of this topology govern how the evidence reviewed below can be read. First, ACE2 sits at a metabolic branch point, so that any phenotype obtained by manipulating it admits two explanations (Section 4.2). Second, Ang-(1–7) acts largely through the Mas receptor to promote nitric oxide (NO) release and antioxidant, anti-inflammatory, antifibrotic, and antihypertrophic signaling,[9,21,22] so the burden of attributing an effect to the axis falls on Mas-selective rather than ACE2-directed tools.

Two further peptides extend this arm: alamandine, which differs from Ang-(1–7) by a single N-terminal residue and signals mainly through the Mas-related G-protein-coupled receptor D (MrgD),[23,24] and the recently described alamandine-(1–5),[25] whose cardiac role is essentially unexplored. They are considered here only insofar as they illuminate protective mechanisms shared with the ACE2/Ang-(1–7)/Mas axis; alamandine is discussed briefly in Section 4.5, and alamandine-(1–5) is placed among the speculative frontiers in Section 8.

3. Physiological versus pathological hypertrophy

Hypertrophic growth begins with the conversion of mechanical and neurohumoral inputs into transcriptional and translational programs.[1] The proximate biophysical stimulus is ventricular wall stress, which rises with intracavitary pressure and chamber radius and falls with wall thickness, so that pressure and volume overload both provoke growth.[26,27] The clinical stakes are set by epidemiology: left ventricular hypertrophy predicts adverse cardiovascular outcomes independently of the load that produced it and is therefore a therapeutic target in its own right.[28,29] Mass alone, however, is a poor discriminator, and the physiological–pathological distinction rests less on the extent of growth than on what accompanies it: interstitial fibrosis, capillary supply, metabolic substrate use, and reversibility.[1,5]

Physiological and pathological hypertrophy are conventionally contrasted as separate cascades, but the difference between them lies in the program rather than in the parts. Their molecular signatures overlap, and shared effectors — including calcineurin/nuclear factor of activated T cells (NFAT) and phosphoinositide 3-kinase (PI3K; p110α)/Akt — can favor either outcome depending on the intensity, duration, and cellular context of activation.[1,4] Two stereotyped configurations are nonetheless useful, and Figure 2 carries the full effector set. Sustained neurohumoral stimulation by Ang II, endothelin-1, and catecholamines acts through Gq/11- and Gs-coupled receptors and the pro-hypertrophic effectors shown in Figure 2, amplified by NADPH-oxidase- and mitochondria-derived reactive oxygen species (ROS); the output is fetal gene reactivation, fibrosis, and contractile dysfunction.[30–32] Growth-factor signaling during development, exercise, and pregnancy instead engages IGF-1/insulin–PI3K(p110α)/Akt/mTOR, complemented by angiogenic, NO/cGMP and AMPK inputs, which expand translational capacity and match angiogenesis to myocyte growth, while tonic inhibition of glycogen synthase kinase 3β (GSK3β) permits controlled growth without maladaptive transcriptional reprogramming.[4,33–35]

What distinguishes the two is therefore not the presence of growth signaling but which effectors are recruited alongside it, and the cleanest available criterion is calcineurin/NFAT coupling, which participates in pathological but not physiological hypertrophy.[36] This criterion carries the analytical weight of the sections that follow. If adaptive and maladaptive growth are defined by their effector profile rather than by their magnitude, a pathway that suppresses calcineurin/NFAT, MAPK, TGF-β/Smad, and ROS signaling while sustaining PI3K/Akt, NO/cGMP, and angiogenic signaling should oppose hypertrophy in one setting and be required for it in the other. That is precisely the behavior of the ACE2/Ang-(1–7)/Mas axis, and the sections that follow test the prediction.

4. The ACE2/Ang-(1–7)/Mas axis in pathological cardiac hypertrophy

The heart contains a complete local Ang-(1–7)-forming system: ACE2 is expressed in cardiomyocytes, endothelial and vascular smooth muscle cells, macrophages and (myo)fibroblasts, and the Mas receptor in cardiomyocytes, fibroblasts and coronary endothelium.[37–42] Table 1 collects the principal genetic and pharmacological evidence; the text below compares those studies rather than recounting them, and reads them against the pathological reference point set by the classical arm.

4.1 The pathological reference point: ACE/Ang II/AT_1 and AT_2

Claims that the protective axis restrains pathological hypertrophy are meaningful only against a defined pathological reference. Ang II acting at AT_1 engages a dense cardiomyocyte network — Gq/PLCβ/IP3/DAG-dependent Ca^{2+} mobilization, protein kinase C, MAPKs, PI3K–Akt–mTOR, calcineurin/NFAT and Ca^{2+}/calmodulin-dependent protein kinase II (CaMKII)/histone deacetylase/MEF2 — amplified by reactive oxygen species derived from the NADPH oxidases NOX2 and NOX4, and it recruits fibroblasts through TGF-β/Smad and ERK1/2 to deposit interstitial and perivascular collagen.[12,43–47]

Whether this myocardial signaling is necessary for hypertrophy is the point at which the two most informative genetic experiments appear to conflict, and the conflict is apparent rather than real: they answer different questions. Cardiomyocyte-restricted AT_1 overexpression produces hypertrophy, fetal gene reactivation, fibrosis and premature heart failure in the absence of systemic hypertension, and therefore establishes sufficiency.[48] Conditional cardiomyocyte-specific deletion of AT_{1A} does not abolish hypertrophy when blood pressure is preserved, and therefore establishes that the same signaling is not necessary and that a substantial component of Ang II-induced growth is transmitted indirectly through vascular AT_1 and afterload.[49] Direct myocardial AT_1

signaling is thus one route among several rather than an obligatory one, and mechanical stretch adds a further, ligand-independent route to the same receptor.[50,51] This distinction governs the remainder of the section: an intervention on the protective axis may act on the myocyte-autonomous route, on the hemodynamic route, or on both, and no study in which blood pressure falls can separate them.

AT_2 is best placed within the protective arm, although the evidence is less uniform. It is re-induced in pathological remodeling and accumulates in interstitial fibroblasts within fibrotic regions, and, unlike AT_1, it couples to phosphatase- and NO/cGMP-dependent pathways that suppress pro-growth MAPK signaling.[52–55] Functional antagonism is supported from both directions: AT_2 blockade augments Ang II-stimulated collagen production and partly abolishes the antihypertrophic effect of AT_1 blockade,[56] whereas selective AT_2 agonists suppress hypertrophy and fibrosis without lowering blood pressure.[53,54] The dissenting observation — constitutive, pro-hypertrophic AT_2 signaling — derives from receptor overexpression in isolated cardiomyocytes and has not been reproduced in vivo, and is best read as context-dependent rather than as a genuine contradiction.[57] One practical corollary is that AT_1 blockade may redirect available Ang II toward re-induced AT_2, in which case part of the structural benefit attributed to AT_1 blockade would itself be counter-regulatory signaling.[8]

4.2 ACE2 as a nodal control point

ACE2 occupies a metabolic branch point, and this single fact sets the limit on what any ACE2 experiment can establish: the enzyme lowers Ang II and raises Ang-(1–7) simultaneously, so a protective phenotype produced by ACE2 manipulation cannot be attributed to Ang-(1–7)/Mas signaling on that evidence alone.

Loss-of-function models are consistent in direction and informative in detail. ACE2 deletion accelerates hypertrophy, fibrosis and systolic dysfunction, and worsens remodeling after aortic constriction, during postnatal development or under Ang II infusion.[58–60] The decisive observation, however, is pharmacological: acute ACE2 inhibition aggravates remodeling in the post-infarction heart,[61] and inhibition with MLN-4760 aggravated hypertrophy and fibrosis in Ren-2 rats through cardiac Ang II accumulation, with blood pressure unchanged.[62] That is direct evidence that part of the benefit of ACE2 reflects removal of Ang II rather than generation of Ang-(1–7).

Gain-of-function studies mirror this pattern: ACE2 overexpression or recombinant ACE2 attenuates hypertrophy, fibrosis and oxidative stress in Ang II infusion, diabetic cardiomyopathy and myocardial infarction.[63–66] That genetic overexpression and delivery of recombinant protein converge on the same phenotype is reassuring, since the two approaches share few off-target liabilities. The most instructive of these is the one furthest from the heart: ACE2 overexpression restricted to the central nervous system reduces Ang II-induced collagen deposition and cardiomyocyte hypertrophy by lowering sympathetic drive, showing that a cardioprotective ACE2 phenotype can be produced without any manipulation of the cardiac peptide balance at all.[67] ACE2 therefore acts through at least three non-exclusive routes — Ang II removal, Ang-(1–7)/Mas gain, and reduced sympathetic outflow — whose relative weight is model-specific and, in most published designs, unmeasured.

The link to the protective peptide is weaker than it is usually presented. Mas blockade has been reported to attenuate or abolish some effects of ACE2 gain of function, but that evidence was obtained largely outside the myocardium,[68,69] and we are aware of no study testing a cardiac antihypertrophic ACE2 phenotype against Mas blockade with the peptides measured in the tissue — which is the experiment on which attribution to Ang-(1–7)/Mas depends. Resolving the branch point requires that design, or ACE2 manipulation on a Mas-null background or with cardiac Ang II clamped. Until then, ACE2 is best described as a nodal control point over the balance of the system rather than as a source of Ang-(1–7).

4.3 Ang-(1–7)/Mas signaling: mechanisms and where models disagree

Evidence obtained by manipulating the peptide and its receptor directly is not subject to the branch-point ambiguity, and it is internally concordant. Ang-(1–7) is an endogenous ligand of Mas, so receptor deletion, receptor blockade and peptide supplementation interrogate one route by three independent means.[21] Ang-(1–7) accumulates in cardiomyocytes bordering ischemic regions, its infusion after myocardial infarction preserves function and reduces myocyte size, and at the cellular level it inhibits cardiomyocyte growth through Mas.[42,70,71] Loss and gain of function agree: Mas-deficient hearts show impaired function, fibrosis and altered Ca^{2+} handling, whereas chronic overproduction of Ang-(1–7) confers resistance to isoproterenol- and Ang II-induced remodeling and protects Ca^{2+} signaling, GSK3β and NFAT from maladaptive change.[41,72,73] Crucially, cardiac-restricted overexpression reproduces against isoproterenol the antifibrotic protection that lifelong systemic overproduction confers across isoproterenol, Ang II and deoxycorticosterone acetate (DOCA)-salt, and does so without a fall in blood pressure — which makes blood-pressure independence one of the better-established properties of the axis.[74–77]

Mechanistically the axis opposes the maladaptive program at several nodes rather than at one: it attenuates ERK1/2, p38, JNK and c-Src phosphorylation;[78–80] inhibits TGF-β1/Smad2;[81] prevents calcineurin/NFAT activation through an NO/cGMP-dependent cascade;[73] raises NO and Akt signaling while limiting nuclear translocation of G-protein-coupled receptor kinase 5, an effect described for a non-peptide Mas agonist;[77] **[FALTA: conferir no PDF original qual referência sustenta o achado de GRK5. A v1 atribui a frase a um agonista não peptídico (CGEN-856S) mas cita `[133]` = de Almeida 2015 (DOCA/disfunção diastólica), que é a ref. 77 desta lista. Ou a citação da v1 estava trocada, ou o estudo de DOCA também mediu GRK5 — os autores precisam confirmar antes da submissão]** engages FOXO3/SOD1/catalase and suppresses NF-κB;[82] and rebalances matrix turnover by raising the matrix metalloproteinases MMP-2 and MMP-9 and lowering their tissue inhibitors TIMP-1 and TIMP-2.[83,84] The list is less important than its shape: the pathways repressed are precisely those that distinguish maladaptive from adaptive growth, which is why the axis behaves as a selector of remodeling program rather than as a generic brake on cardiomyocyte size.

Concordance in direction, however, conceals three systematic disagreements between models. The first is between endpoints. Studies that report fibrosis, diastolic function and fetal gene expression describe consistent protection, whereas studies whose primary endpoint is ventricular mass report smaller and less reliable effects. The dissociation also appears within single studies, most clearly in subtotal nephrectomy (Section 4.4), where the same compound reduced fibrosis and left mass untouched; how general it is cannot be settled from the present literature, since we are not aware of any study comparing mass against tissue endpoints systematically across models. Read as a defect, this is inconsistency; read as a result, it is the central finding of this review, because it says that the axis governs the character of remodeling more reliably than its magnitude. The second is between stimuli. Protection is best documented against neurohumoral drive — Ang II infusion, isoproterenol, DOCA-salt, genetic hypertension — where the axis directly opposes the initiating signal, while the evidence against mechanically initiated pressure overload is thinner and unevenly distributed along the axis. At the level of ACE2 it is direct: deletion accelerates dysfunction after aortic constriction,[58] and ACE2 activators protect in aortic banding.[85] At the level of the ligand and its receptor it is not: protection after transverse aortic constriction has been shown for alamandine, through MrgD rather than Mas,[86] and the only direct manipulation of Ang-(1–7) against a mechanical stimulus is against the volume overload of an aortocaval fistula (Section 5), not against pressure overload. Whether Ang-(1–7)/Mas itself restrains hypertrophy that is initiated by pressure rather than humorally is therefore not yet established, and it is a question that bears directly on aortic stenosis and on hypertensive hearts in which pressure is controlled. The third is between preparations: cellular mechanisms are established largely in neonatal rodent cardiomyocytes, whereas the phenotypes they explain are adult, and hypertrophy is quantified variously as myocyte cross-sectional area or as mass normalized to body weight or tibial length — measures that respond differently to interstitial change and make effect sizes only loosely comparable. None of the three disagreements is resolved by adding studies of the same design; each requires a design the field has not standardized.

4.4 Pharmacological and genetic activation: promise and inconsistencies

Pharmacological activation broadly reproduces the genetic findings, and the places where it does not are the most informative part of the literature. The ACE2 activator xanthenone (XNT) attenuates hypertension- and diabetes-induced fibrosis and hypertrophy, partly through ERK1/2,[87,88] and diminazene aceturate (DIZE) prevents or partly reverses hypertrophy and improves diastolic function across four etiologies that share little except a raised classical arm — myocardial infarction, two-kidney one-clip (2K1C) renovascular hypertension, the *db/db* diabetic mouse and aortic banding (Table 1).[85,89–92] In subtotal nephrectomy, by contrast, the same compound reduced interstitial and perivascular fibrosis while leaving left ventricular hypertrophy unchanged.[93] That result is usually cited as a negative finding; read against the endpoint dissociation described above, it is the expected one, and it locates the boundary of the drug effect: antifibrotic and antihypertrophic actions are separable, and which of the two appears depends on etiology.

Two cautions limit how far these compounds can carry an argument. First, as with any small-molecule activator, specificity cannot be inferred from a phenotype alone: neither agent is a receptor-selective tool, and a benefit obtained pharmacologically is therefore weaker evidence for the axis than an equivalent benefit obtained genetically. Second, dose, duration and route differ across the studies cited above, and we are not aware of a head-to-head comparison across etiologies with common endpoints, so apparent inconsistency and genuine model dependence cannot presently be told apart.

The genetic counterpart of these interventions is more specific but narrower: overexpression of ACE2 and of Ang-(1–7) protects in overlapping but not identical models, and we are not aware of the two having been compared side by side in one laboratory, which is the simplest experiment that would apportion the benefit of ACE2 between peptide removal and peptide generation. A useful way to grade the section is by convergence. Where genetic and pharmacological approaches agree — Ang II infusion, diabetic cardiomyopathy, myocardial infarction — the antifibrotic and function-preserving effect of activating the axis can be treated as established. Where only part of the axis has been manipulated — mechanically initiated pressure overload, where the evidence stops at ACE2, and renal disease — the effect is emerging rather than established. And regression of existing hypertrophy, as opposed to prevention of its development, rests on very few studies and should be treated as speculative. On the translational side, an orally active Ang-(1–7) inclusion compound produces cardiovascular effects comparable to exercise training in spontaneously hypertensive rats (SHR), which suggests that the short half-life of the peptide is a surmountable formulation problem rather than a barrier in principle.[94]

4.5 Alamandine/MrgD: a parallel protective arm

A structurally adjacent peptide reproduces much of this biology through a different receptor. Alamandine acts mainly at MrgD: its genetic deletion causes dilated cardiomyopathy under basal conditions, and alamandine prevents Ang II-induced hypertrophy in vitro and attenuates remodeling after transverse aortic constriction, in spontaneous hypertension, after myocardial infarction and after neonatal hyperoxia, often independently of blood pressure.[86,95–99] Its antihypertrophic effect is blocked by the MrgD antagonist D-Pro7-Ang-(1–7) but not by the Mas antagonist A-779, and converges on AMPK activation, a node at which long-term activation is itself antihypertrophic.[96,100,101] This pharmacological separation has a consequence for the preceding sections that is rarely stated: because alamandine signaling is insensitive to A-779, protection that persists under Mas blockade cannot be assumed to be non-peptidergic, and the parallel arm remains a candidate explanation for the incomplete blockade seen in ACE2 studies. Full dissection of alamandine signaling lies beyond the scope of a review centered on Ang-(1–7), but the convergence of the two peptides on AMPK argues that the protective arm is better described as a set of overlapping mechanisms than as a single pathway. A further product of this arm, alamandine-(1–5), has recently been identified; its cardiac role is unexplored, and it is treated here only as a boundary of the current map.[25]

DECISÃO DO ORIENTADOR: peso do Ala-(1–5). Mantive o mínimo defensável — uma frase, sem afirmação cardíaca (não existe dado cardíaco publicado), enquadrada como fronteira do que se sabe. O editor nomeou o peptídeo entre os excessos; o Robson é coautor e a descoberta é do grupo. Opções: (a) manter como está; (b) cortar daqui e deixar só a menção da §2 + a faixa "speculative" da §8; (c) ampliar para duas ou três frases, assumindo o risco com o editor. QUEM DECIDE: o Carlos, como orientador, e vale ouvir o Robson antes, já que a descoberta é do grupo dele e ele é coautor.

5. The axis in right ventricular hypertrophy

Right ventricular hypertrophy (RVH) arises most often from pulmonary hypertension, and it is not left ventricular hypertrophy relocated to a thinner chamber. Its maladaptive transition is defined by features that are either without a close left-sided counterpart or far more prominent than on the left: a glycolytic shift driven by hypoxia-inducible factor 1α (HIF-1α) and resembling a Warburg phenotype, mitochondrial dysfunction, severe capillary rarefaction, and an early loss of ventricular–arterial coupling.[102–105] Whether conclusions established in the left ventricle transfer to the right is therefore a real question, not a formality.

The available answer is affirmative but narrow (Table 1). ACE2 protein is present across right ventricular myocardium, epicardium and endocardium, which is consistent with a local axis in the chamber;[106] whether Mas is expressed in right ventricular myocardium has not, to our knowledge, been reported, so the local axis is inferred from its upstream enzyme alone. Classical RAS components rise in RVH and track disease progression,[107,108] and activation of the protective arm reduces RVH and fibrosis across models: ACE2 activation with XNT or DIZE in monocrotaline-induced pulmonary hypertension shifts the local RAS toward the protective profile and raises Mas and IL-10 expression,[109,110] while lentiviral Ang-(1–7) overexpression lowers right ventricular systolic pressure, hypertrophy and fibrosis in an A-779-sensitive manner.[111]

The interpretive difficulty is that most of these models are pulmonary vascular. An intervention that lowers pulmonary arterial pressure also unloads the right ventricle, so regression of hypertrophy need not be a myocardial effect at all. Only two designs separate the two: recombinant ACE2 improved right ventricular systolic and diastolic function under surgically fixed afterload,[106] and Ang-(1–7)-overproducing rats were protected against the volume overload of an aortocaval fistula.[112] To our knowledge that is the whole of the load-independent evidence. Beyond it, we are aware of no right-ventricle-restricted or cardiomyocyte-restricted genetic model, no characterization of the right ventricle of Mas-deficient animals, and no study asking whether the axis modifies the metabolic and microvascular changes that specifically define right ventricular failure. The axis is protective in the right ventricle; whether it protects the right ventricle for right-ventricular reasons is untested.

6. The ACE2/Ang-(1–7)/Mas axis in physiological hypertrophy

6.1 Exercise

Aerobic training produces reversible left ventricular hypertrophy with balanced growth, preserved function and minimal fibrosis,[1] and three loss-of-function experiments indicate that the ACE2/Ang-(1–7)/Mas axis is required for this adaptation rather than merely coincident with it. They do not, however, agree on what the axis is required for (Table 1). ACE2-deficient mice fail to gain heart and left ventricular mass during voluntary running, so the growth response itself is lost, while training upregulates left ventricular Mas in their wild-type littermates.[113] Mas-deficient mice, by contrast, hypertrophy normally but accumulate collagen I and III and lose the training-induced rise in left ventricular Ang-(1–7): magnitude is intact and quality is not.[114] Acute Mas blockade with A-779 diverges from both, preventing exercise-induced cardiomyocyte hypertrophy outright and leaving myofibrillar atrophy and abnormal mitochondria.[115] The discrepancy between constitutive deletion and acute antagonism is most economically explained by developmental compensation in the knockout, and it warns against reading either result as the definitive measure of Mas dependence.

Training also shifts the cardiac RAS toward the protective arm, and the shift is graded rather than uniform. Swimming lowers cardiac ACE activity, expression and Ang II while raising ACE2 activity and Ang-(1–7), with concordant microRNA changes — increased miR-27a/b targeting ACE, decreased miR-143 targeting ACE2 — that place part of the regulation post-transcriptionally.[116] Intensity dissociates the two readouts once more: swimming protocols of different volume and intensity produced comparable hypertrophy, yet only moderate intensity raised Mas expression and Akt activation, leaving AT_1 and ERK unchanged.[117] Baseline state, however, discriminates better than dose, and the pattern is best read as a single principle: the axis is recruited in proportion to how far the classical arm is already engaged. Three settings test that principle from different baselines and agree: the training-induced rise in left ventricular Ang-(1–7) and Mas is selective for hypertensive animals and absent in normotensive controls,[118] prolonged training in established hypertension raises ACE2 and Mas while lowering ACE and AT_1,[119] and intense interval training rebalances the local RAS in diet-induced hypertrophy.[120] On this reading the axis looks less like a constitutive component of exercise signaling than like a corrective one, engaged where there is an imbalance to correct.

Downstream, the two converge on the same effectors: PI3K/Akt is the accepted mediator of physiological growth,[121,122] its loss blunts the response to exercise while exaggerating the response to pathological stimuli,[123] inhibitory phosphorylation of GSK3 permits controlled growth,[124] and AMP-activated protein kinase (AMPK) is activated both by training[125] and, within the parallel protective arm, by alamandine[96] (Figure 3). Convergence, however, is the weakest form of the argument here. Outside the three loss-of-function studies above, the exercise literature measures RAS components alongside hypertrophy rather than testing their necessity; and the orally active Ang-(1–7) compound that reproduces exercise-like cardiovascular effects does so in SHR,[94] that is, by attenuating pathological growth, so it supports the pharmacological viability of the ligand and not its sufficiency for physiological growth.

6.2 Pregnancy

Pregnancy is the more instructive of the two physiological models because its stimulus is overtly pro-hypertrophic: circulating renin, angiotensinogen and Ang II all rise, yet the maternal heart grows reversibly and without loss of function.[126,127] Gestation does not merely tolerate Ang II; it protects the heart against its antiangiogenic and fibrogenic effects, and the cardiac gene programs for function, angiogenesis and apoptosis that Ang II induces during pregnancy run opposite to those it induces outside it.[128] Mas is implicated directly: A-779 or genetic Mas deletion attenuates pregnancy-induced cardiomyocyte hypertrophy and increases left ventricular collagen III.[129] This is not the exercise result: there, Mas deletion left growth magnitude intact and degraded only its quality, whereas in pregnancy both are affected — either because gestational growth depends more heavily on the axis, or because the volume-overload stimulus is weaker and therefore less able to proceed without it.

An underappreciated inconsistency constrains interpretation. In pregnant rats and mice these Mas-dependent effects occur with no measurable change in circulating Ang II or Ang-(1–7), whereas in pregnant women both peptides rise.[130,131] **[FALTA: conferir a espécie da ref. 130 (Brosnihan, *Braz J Med Biol Res* 2004). Se o dado for de rata, ela apoia a cláusula anterior (roedores) e não a das gestantes humanas, que ficaria sustentada só pela ref. 131 (Merrill 2002)]** The rodent data therefore point to local cardiac Mas signaling rather than systemic ligand availability, but that local ligand has never been identified, and the human pattern would, if anything, predict the opposite division of labor: rodent-to-human translation here is an open question rather than a detail. ACE2 itself has a demonstrable gestational role, but it is extracardiac — ACE2-deficient dams show impaired gestational weight gain and fetal growth restriction[132] — so the enzyme cannot be manipulated in pregnancy without confounding maternal and fetal phenotypes, and no equivalent of the exercise ACE2-knockout experiment exists here.

Mechanistically, pregnancy engages the PI3K/Akt/eNOS/GSK3 nodes that Ang-(1–7)/Mas also engages in cardiomyocytes,[124,133,134] with demonstrated ERK1/2–PI3K/Akt cross-talk during gestational growth.[135] That overlap is consistent with a shared program but does not establish causal engagement of the axis, and the in vivo case rests on a single study.[129] A further gap is that the reversibility which defines gestational hypertrophy — its regression after delivery — is the readout the quality-versus-magnitude claim most directly predicts, and, to our knowledge, it has not been examined under manipulation of the axis. The pregnancy evidence should therefore be labeled suggestive rather than established.

6.3 A unifying interpretation: quality, not magnitude

Read together, the exercise and pathological literatures support a single interpretation. The most informative contrast is genetic: Mas deletion preserves the magnitude of exercise-induced hypertrophy but degrades its quality, producing collagen I/III accumulation,[114] whereas ACE2 deletion abolishes the hypertrophic response altogether.[113] Acute Mas blockade sits with neither, removing the growth response itself and leaving ultrastructural degeneration[115] — the discrepancy discussed in Section 6.1, which the interpretation below must survive rather than absorb. The dissociation between the two genetic models implies that Mas signaling governs the character of remodeling, while ACE2 additionally gates the growth signal itself — consistent with the upstream, branch-point position of ACE2. Reinforcing the point, the effectors repressed by Mas (calcineurin/NFAT, MAPK cascades, TGF-β/Smad and ROS) are those that define the maladaptive program, whereas the pathways it sustains (PI3K/Akt, NO/cGMP, FOXO3-dependent antioxidant defense, and angiogenesis matched to myocyte growth) are those that characterize adaptive growth.[36,73,78,82]

The pregnancy and right ventricular literatures fit the same shape without adding independent weight: in pregnancy Mas loss again shifts remodeling toward collagen deposition,[129] and in the right ventricle quality endpoints have been measured — fibrosis, inflammation, systolic and diastolic function — but never dissociated from magnitude within a single study, which is the comparison the framework needs.

This reframing resolves the apparent paradox. A pathway that biases the myocardium away from fibrosis, oxidative stress and contractile dysfunction and toward structurally and energetically favorable remodeling will restrain hypertrophy where the prevailing program is maladaptive (hypertension, chronic neurohumoral drive) yet prove necessary where the program is adaptive (exercise, pregnancy). The axis is therefore better described as a regulator of remodeling quality than as a generic antihypertrophic brake — an interpretation coherent with the available data but still largely inferential (Section 8). It is, however, falsifiable: it predicts that graded Mas modulation should move fibrosis, oxidative and angiogenic endpoints while leaving myocyte cross-sectional area comparatively undisturbed, and that any intervention which suppresses growth magnitude without improving remodeling quality is acting elsewhere.

7. Early-life programming of the axis (DOHaD)

DECISÃO DO ORIENTADOR: manter esta seção ou cortá-la por inteiro — é a decisão 5.5 do plano de corte. O editor listou DOHaD entre os excessos periféricos; ela sobrevive aqui só porque fala do eixo e é linha do grupo. Corte total libera ~250 palavras para a §4, onde rendem mais. Se cortar, remover também (a) a menção "the developmental programming of Section 7" no tier *Emerging* da §8 e (b) renumerar §8→§7 e §9→§8, e (c) as refs 136–141 e 143–145, que só aparecem aqui. ATENÇÃO: a ref. 142 (Rueda-Clausen 2011) NÃO morre com o corte — a §8 a cita na frase sobre dependência de sexo, e removê-la deixa citação órfã. QUEM DECIDE: o Carlos, como orientador. Corte esta seção só com o OK dele — é linha do grupo, e o custo de manter é ficar 583 palavras acima do alvo.

The axis also operates before the heart is ever challenged. Cardiac vulnerability is set by the developmental transition from hyperplastic to hypertrophic growth around the second postnatal week;[136] hypoxia and undernutrition within this window reduce cardiomyocyte proliferation, force premature terminal differentiation, and limit cardiomyocyte endowment,[137–139] leaving a myocardium with less reserve for later adaptive growth. Programmed phenotypes are not RAS-neutral: they carry a shifted balance — higher Ang II, altered AT_1/AT_2 and Mas, reduced MrgD — and are frequently sex-dependent, with male-predominant left ventricular hypertrophy.[140–142] Prepuberty behaves as a second sensitive window, in which nutritional insults produce hypertrophy, fibrosis, and increased AT_1/AT_2 expression.[143]

The observations that bear directly on our argument are interventional, and the two windows differ in a way that matters (Table 1). Maternal treatment of pregnant SHR with DIZE or Ang-(1–7) attenuated the development of hypertension in the offspring as well as reducing their cardiomyocyte hypertrophy and fibrosis in adulthood, so the structural benefit there cannot be separated from the pressure benefit.[144] Prepubertal Ang-(1–7), by contrast, reduced left ventricular wall thickness, cardiomyocyte hypertrophy and fibrosis in adulthood without affecting systolic blood pressure, while preserving Mas expression and lowering ERK1/2 phosphorylation and oxidative stress.[145] It is the second, load-independent result that carries the argument. Its signature is the one seen in adult hearts — fibrotic and oxidative rather than purely dimensional — but installed developmentally and durable, which suggests that early modulation of the axis sets, rather than merely modifies, the remodeling program the heart will later run. The interest here is conceptual: remodeling quality may be a programmable property, and the window in which the axis is most tractable may precede the disease it is meant to prevent. The claim remains emerging rather than established — few studies, a single strain, no human data.

8. Limitations and future perspectives

The evidence on this axis falls into three tiers, and much of the apparent disagreement in the literature comes from treating them as one.

Established. That the axis opposes fibrosis and oxidative stress in pathological remodeling is supported by concordant genetic and pharmacological evidence, independently of blood pressure, in the etiologies where both approaches have been applied — Ang II excess, diabetic cardiomyopathy and myocardial infarction;[8,9] so is the finding that ACE2 loss aggravates and ACE2 gain attenuates remodeling.[62] These are the claims that survive a change of model. Effects on ventricular mass itself belong one tier lower, for the reasons given in Section 4.3.

Emerging. The requirement of the axis for physiological growth rests on far fewer studies: ACE2 deletion abolishes the exercise response,[113] Mas deletion preserves its magnitude and degrades its quality,[114] and acute Mas blockade abolishes the growth response itself[115] — a discrepancy discussed in Section 6.1 and not yet resolved. The pregnancy evidence is thinner still.[129] Alamandine/MrgD as a parallel arm[95,96] and the developmental programming of Section 7 belong in this tier.

Speculative. The central claim of this review — that the axis sets the quality rather than the magnitude of remodeling — is coherent with the data but has never been tested directly. Alamandine-(1–5)[25] has no cardiac data of any kind. And the therapeutic proposition is wholly untested: we are aware of no clinical trial evaluating activation of the protective arm against cardiac hypertrophy in humans.

Four gaps do most of the constraining. Attribution comes first: the branch-point problem of Section 4.2 means that no ACE2 manipulation — genetic, pharmacological, or recombinant — can by itself assign a phenotype to Ang-(1–7)/Mas;[62] A-779 narrows this ambiguity without closing it, and conditional, cell-type-specific models and receptor-selective or biased agonists are the missing tools. Second, model dependence: XNT and DIZE are small-molecule activators whose specificity cannot be inferred from a phenotype, and the antifibrotic action of DIZE dissociates from its antihypertrophic action in subtotal nephrectomy,[93] so standardized models, endpoints and dosing are a precondition for comparing pharmacological results across etiologies at all. Third, the ligand driving local signaling is unidentified: in pregnant rodents Mas-dependent adaptation proceeds without measurable change in circulating Ang II or Ang-(1–7), whereas both rise in pregnant women[129–131] — here the rodent-to-human step is not merely unvalidated, it points the wrong way. Fourth, sex: pathological and exercise work is predominantly male,[146] the pregnancy literature is necessarily female, and programmed phenotypes are themselves sex-dependent,[142] leaving the generality of every claim above untested in one sex or the other.

Two lines of work would move claims between tiers rather than add to them. Single-cell and spatial transcriptomic dissection of myocyte and non-myocyte compartments under graded Mas modulation, with growth, fibrosis, and angiogenesis quantified separately in the same hearts, would test the quality hypothesis directly instead of inferring it — in the studies reviewed here, remodeling quality is seldom reported as an endpoint in its own right. And the pharmacology of the ligand needs settling: orally active Ang-(1–7) formulations show encouraging preclinical cardiac effects,[94] but their pharmacokinetics, safety, and efficacy are not characterized to the standard a first-in-human trial requires.

9. Concluding remarks

The ACE2/Ang-(1–7)/Mas axis looks paradoxical only when it is read as an antihypertrophic system. It restrains cardiac growth under hypertension, Ang II excess and myocardial infarction, yet exercise-induced growth fails without ACE2 and loses its adaptive character without Mas. Both observations follow from one function: the axis sets the quality of remodeling, not its magnitude. Where the prevailing program is maladaptive, that function reads as a brake; where it is adaptive, as a requirement.

Read this way, the axis is not a competitor to classical RAS blockade, which lowers load and with it growth, but a different kind of target: it acts on what growth becomes. The therapeutic goal it implies is not to suppress hypertrophy — which would also suppress the growth that exercise and pregnancy demand — but to redirect remodeling: to oppose its fibrotic, oxidative, and microvascular components while preserving, or improving, the growth itself. That goal is at present a hypothesis with substantial preclinical support and no clinical test of any kind. What the field needs is therefore not further demonstrations that the axis is protective, but experiments designed to measure remodeling quality directly and to separate it from mass — the measurement on which this framework stands or falls.

Statements and Declarations

Ethical considerations

Not applicable. This review did not involve new studies on human participants, human data, human tissue, or animals.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Declaration of conflicting interests

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Data availability

Data sharing is not applicable to this article, as no new data were created or analyzed.

Author contributions

[FALTA: seção Author contributions por extenso, com as iniciais de cada autor e o papel de cada um. Na v2 isto estava em rascunho entre colchetes — `[To be confirmed by the authors, e.g.: C.H.C. and M.M.L.T. conceptualized the review and drafted the manuscript and contributed equally; R.A.S.S. and J.B.R.D. critically revised the manuscript for important intellectual content; all authors read and approved the final version.]` **— e sumiu na v3. Só os autores sabem quem fez o quê; o rascunho da v2 serve de ponto de partida, não de resposta. Observar que esta seção é peça distinta da nota de rodapé de contribuição igualitária da página de rosto, e que ambas precisam existir]**

Supplementary material

[FALTA: confirmar se haverá material suplementar. Se não houver, a declaração é dispensável; se houver, ela é obrigatória e nenhuma das três versões a traz]

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[FALTA: completar os DOIs ausentes nas refs 14, 18, 19, 20, 21, 27, 48, 68, 69, 72, 74, 91, 108 e 136 (as outras 132 têm DOI). Buscar no CrossRef e colar o registrado — DOI preenchido de memória é tão reprovável quanto referência inventada. Se algum realmente não tiver DOI registrado (as mais antigas, a 136 é de 1984), deixar sem, e aí a ausência fica consistente]

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Table 1. Selected genetic and pharmacological evidence on the ACE2/Ang-(1–7)/Mas axis in cardiac hypertrophy, with the classical-arm reference point against which it is read.

Model and setting	Intervention (direction of change)	Main cardiac outcome	What the finding supports or challenges	Source
Classical-arm reference point				
Cardiomyocyte-restricted AT_1 overexpression; normotensive mouse	Cell-autonomous gain of classical-arm signaling	Hypertrophy, fetal gene reactivation, fibrosis and early heart failure in the absence of systemic hypertension	Supports **sufficiency** of direct myocardial AT_1 signaling. Fixes the maladaptive reference point without which "restraint" by the protective axis has no measurable counterpart	48

Model and setting	Intervention (direction of change)	Main cardiac outcome	What the finding supports or challenges	Source
Cardiomyocyte-specific AT_{1A} deletion; blood pressure preserved	Cell-autonomous loss of classical-arm signaling	Hypertrophy not abolished	Challenges **necessity** of cardiomyocyte AT_1, and indicates that a substantial part of Ang II-driven growth is afterload-mediated. Read against the row above, the two experiments are not in conflict: they answer sufficiency and necessity, and only their conflation produces an apparent contradiction	49
ACE2: loss of function				
Global ACE2 knockout; pressure overload, postnatal development, Ang II infusion	Genetic loss of function	Accelerated hypertrophy, perivascular and interstitial fibrosis, systolic dysfunction; worse remodeling after each stressor	Supports ACE2 as necessary for normal tolerance of load. Does **not** discriminate Ang II removal from Ang-(1–7) generation (branch point, Section 4.2) — the central interpretive limit of the whole genetic literature	58–60
Pharmacological ACE2 inhibition; Ren-2 hypertensive rats (MLN-4760) and post-infarction heart	Acute pharmacological loss of function	Aggravated hypertrophy and fibrosis with cardiac Ang II accumulation, independent of blood pressure	Directly challenges an exclusively Ang-(1–7)/Mas reading of ACE2 benefit: at least part of that benefit is Ang II removal, not peptide generation	61, 62
ACE2: gain of function				
ACE2 overexpression (systemic lentiviral, cardiac, central) or recombinant ACE2; Ang II infusion, diabetic cardiomyopathy, myocardial infarction	Genetic or protein gain of function	Reduced hypertrophy, fibrosis and oxidative stress across etiologies; central overexpression reduces Ang II-induced remodeling by lowering sympathetic drive	Supports translational plausibility of raising axis tone. The central-overexpression result also shows that part of the structural benefit is neurogenic rather than myocardial, which complicates attribution of any whole-animal phenotype to local cardiac signaling	63–67
XNT (ACE2 activator); hypertension and diabetes models	Pharmacological gain of function	Reduced fibrosis and hypertrophy, in part ERK1/2-dependent	Supports the axis as a druggable target; specificity of a small-molecule activator cannot be inferred from the phenotype it produces, so these data corroborate rather than establish the mechanism	87, 88
DIZE (ACE2 activator); myocardial infarction, 2K1C, *db/db*, aortic banding	Pharmacological gain of function	Improved diastolic function after infarction; hypertrophy prevented or partly reversed	Supports reproducibility of the pharmacological effect across renovascular, metabolic and pressure-overload etiologies	85, 89–92

Model and setting	Intervention (direction of change)	Main cardiac outcome	What the finding supports or challenges	Source
DIZE; subtotal (5/6) nephrectomy	Same intervention, different etiology	Interstitial and perivascular fibrosis reduced; left ventricular hypertrophy unchanged	Challenges the assumption that the antifibrotic and antihypertrophic actions are coupled. Compared with the row immediately above, this is the clearest demonstration of model dependence in the pharmacological literature and the strongest argument for standardized endpoints across etiologies before extrapolating from any single model	93
Ang-(1–7)/Mas: gain of function				
Lifetime circulating Ang-(1–7) overproduction, TGR(A1-7)3292 rat; Ang-(1–7) delivery in Ang II-induced remodeling	Genetic and pharmacological gain of function	Less hypertrophy and fibrosis, independent of blood pressure; Ca^{2+} handling, GSK3β and NFAT protected from maladaptive change	Supports a ligand-specific, blood-pressure-independent action of the axis. This is the best-established element of the evidence base, because the effect survives hemodynamic control and is reproduced across stimuli	72, 75, 76
Cardiac-specific Ang-(1–7) overexpression; isoproterenol injury	Tissue-restricted gain of function	Protection against isoproterenol-induced fibrosis, independent of blood pressure	Supports a local myocardial action and excludes systemic hemodynamics as the explanation — the tissue-restricted counterpart of the row above	74
Loss of function in pathological and physiological settings: the informative dissociation				
Mas knockout; basal conditions	Genetic loss of function	Impaired function, fibrosis, altered Ca^{2+} handling	Supports a tonic requirement for Mas signaling even without an imposed stressor	41
Mas knockout; exercise training	Genetic loss of function	Magnitude of exercise-induced growth preserved, but collagen I/III accumulation	Supports the central claim that the axis regulates the **quality** rather than the magnitude of remodeling: growth persists, its character does not	114
Mas blockade with A-779; exercise training	Pharmacological loss of function	Exercise-induced cardiomyocyte hypertrophy prevented, with collagen deposition and ultrastructural degeneration	Partly challenges the row above: pharmacological blockade removes the growth itself, whereas constitutive deletion does not. Developmental compensation in the knockout, and dose or off-target effects of A-779, remain unseparated — a gap that no published study resolves	115

Model and setting	Intervention (direction of change)	Main cardiac outcome	What the finding supports or challenges	Source
ACE2 knockout; voluntary running	Genetic loss of function one step upstream	Exercise-induced increase in heart and left ventricular mass fails to occur	Supports ACE2 gating the growth signal itself, in addition to Mas governing its character; consistent with the branch-point position of the enzyme and with the discordance between the two rows above	113
Mas deletion or A-779; pregnancy	Genetic and pharmacological loss of function	Pregnancy-induced hypertrophy attenuated; collagen III increased	Extends the requirement for the axis to a second, independent physiological stimulus, so the exercise result is not stimulus-specific	129
Parallel arm and right ventricle				
Alamandine in TAC, spontaneous hypertension and myocardial infarction; MrgD knockout	Gain and loss of function in the parallel arm	Reduced hypertrophy and fibrosis, AMPK-dependent; MrgD deletion produces dilated cardiomyopathy	Supports shared protective mechanisms across the counter-regulatory arm, with AMPK as a convergence node. Being Mas-independent, it corroborates the general logic of the protective arm without testing the ACE2/Ang-(1–7)/Mas axis itself	86, 95, 96
Recombinant ACE2, XNT, DIZE or Ang-(1–7) augmentation; pulmonary hypertension and right ventricular pressure overload	Pharmacological and protein gain of function	Improved right ventricular function, reduced hypertrophy and fibrosis, local RAS shifted toward the protective profile; Mas dependence tested, and demonstrated, only for the Ang-(1–7) overexpression arm[111]	Supports transfer of the left ventricular conclusions to the right ventricle, but the evidence base is far smaller and no study addresses whether the right ventricle has determinants of its own. Transferability is therefore supported, not established	106, 109–111
Early-life programming				
Ang-(1–7) or DIZE given to pregnant SHR, or Ang-(1–7) given to prepubertal offspring	Pharmacological gain of function within a developmental window	Reduced hypertrophy and fibrosis in adult offspring; maternal treatment also attenuated the development of hypertension,[144] whereas the prepubertal effect occurred without a change in systolic blood pressure[145]	Emerging rather than established, and only the prepubertal arm is load-independent: supports developmental programming of the axis, but rests on few studies in a single strain, with no independent replication and no human counterpart	144, 145

2K1C, two-kidney one-clip; AMPK, AMP-activated protein kinase; Ang, angiotensin; AT_1, angiotensin type 1 receptor; DIZE, diminazene aceturate; GSK3β, glycogen synthase kinase 3 beta; MrgD, Mas-related G-protein-coupled receptor D; NFAT, nuclear factor of activated T cells; RAS, renin–angiotensin system; SHR, spontaneously hypertensive rats; TAC, transverse aortic constriction; XNT, xanthenone.

Figure legends

NOTA DA INTEGRAÇÃO: legendas reaproveitadas na íntegra da v2, que é a única versão que as traz completas com a lista de abreviaturas. Não entram no orçamento de palavras do corpo.

Figure 1. Schematic representation of the classical and counter-regulatory renin–angiotensin system pathways. ACE, angiotensin-converting enzyme; ACE2, angiotensin-converting enzyme 2; Ala-(1–5), alamandine-(1–5); Ang I, angiotensin I; Ang II, angiotensin II; Ang III, angiotensin III; Ang A, angiotensin A; Ang-(1–5), angiotensin-(1–5); Ang-(1–7), angiotensin-(1–7); Ang-(1–9), angiotensin-(1–9); APA, aminopeptidase A; AT1, angiotensin type 1 receptor; AT2, angiotensin type 2 receptor; DCase, decarboxylase; Mas, Mas receptor; MrgD, Mas-related G-protein-coupled receptor D; NEP, neprilysin; PEP, prolyl endopeptidase.

Figure 2. Signaling pathways involved in physiological and pathological cardiac hypertrophy. Pathological hypertrophy is primarily driven by chronic neurohumoral stimulation (angiotensin II, endothelin-1, catecholamines) acting through Gq- and Gs-coupled receptors, promoting calcium dysregulation, oxidative stress, and activation of calcineurin/NFAT, CaMKII, PKC, and MAPK cascades (ERK1/2, JNK, p38), leading to maladaptive gene expression, fibrosis, and contractile dysfunction. Physiological hypertrophy is induced by exercise, IGF-1, mechanical loading, and angiogenic factors, which activate the PI3K/Akt pathway and its downstream effectors (eNOS, NO, cGMP/PKG, mTOR, AMPK), promoting coordinated growth, angiogenesis, efficient metabolism, preserved calcium handling, and maintained function. Solid arrows indicate activation; dashed arrows indicate inhibitory phosphorylation. AC, adenylyl cyclase; AMPK, AMP-activated protein kinase; Akt, protein kinase B; CaMKII, Ca2+/calmodulin-dependent protein kinase II; cAMP, cyclic adenosine monophosphate; cGMP, cyclic guanosine monophosphate; eNOS, endothelial nitric oxide synthase; ERK1/2, extracellular signal-regulated kinases 1/2; GC, guanylyl cyclase; GSK3β, glycogen synthase kinase 3 beta; HDAC, histone deacetylase; IGF-1, insulin-like growth factor-1; IP3, inositol 1,4,5-trisphosphate; IRS1/2, insulin receptor substrate 1/2; JNK, c-Jun N-terminal kinase; LKB1, liver kinase B1; MAPKK, mitogen-activated protein kinase kinase; mTOR, mechanistic target of rapamycin; NFAT, nuclear factor of activated T cells; NO, nitric oxide; NOXs, NADPH oxidases; PI3K, phosphoinositide 3-kinase; PKA, protein kinase A; PKC, protein kinase C; PKG, protein kinase G; PLC, phospholipase C; ROS, reactive oxygen species; VEGF, vascular endothelial growth factor.

Figure 3. Schematic representation of the signaling pathways associated with physiological cardiac hypertrophy induced by physical exercise and by activation of the protective arm of the renin–angiotensin system, showing the convergence of intracellular pathways involved in adaptive cardiac growth. Solid arrows indicate activation; dashed arrows indicate inhibitory phosphorylation. PI3K, phosphatidylinositol 3-kinase; Akt, protein kinase B; eNOS, endothelial nitric oxide synthase; NO, nitric oxide; GC, guanylate cyclase; cGMP, cyclic guanosine monophosphate; PKG, protein kinase G; GSK3β, glycogen synthase kinase 3 beta; mTOR, mechanistic target of rapamycin; ERK1/2, extracellular signal-regulated kinase 1/2; AMPK, AMP-activated protein kinase; IGF1, insulin-like growth factor 1.